

## QUANTITATIVE ECONOMICS - FALL 2024

Answer any FIVE questions.

1. Consider an independent random sample of data size  $n$  drawn from the continuous distribution of  $X \sim (\mu_x, \sigma_x^2)$ , i.e.,  $\{x_i\}_{i=1}^n$ . Suppose for whatever reason, we do not like the first observation and we propose the following estimator of  $\mu_x$ :

$$\tilde{x} = \frac{x_2 + x_3 + \cdots + x_n}{n}$$

With this information, answer the following (show your work):

- What is the expected value of  $\tilde{x}$ ?
  - What is the bias of  $\tilde{x}$ ?
  - What is the variance of  $\tilde{x}$ ?
  - What is the mean square error of  $\tilde{x}$ ?
  - What happens to the results in parts (a-d) when the sample size  $n$  tends to infinity? What can be said about this estimator in this scenario?
2. Consider the model

$$y = X\beta + e$$

where  $E(e|X) = 0$  and  $E(e^2|X) = \Omega$ , where  $\Omega$  is a known variance-covariance matrix. Our goal is to estimate  $\beta$  via minimizing the objective function  $\tilde{e}'\Omega^{-1}\tilde{e}$ , where  $\tilde{e} = y - X\tilde{\beta}$ . With this information

- Derive the estimator  $\tilde{\beta}$  of  $\beta$ .
  - Show that the estimator from part (a) is unbiased.
  - Derive the variance of the estimator from part (a).
3. Consider the following data generating process

$$Y_t = c + \phi_2 Y_{t-2} + \varepsilon_t$$

where  $\varepsilon_t$  is a white noise process. Assuming  $0 < \phi_2 < 1$ , answer the following:

- Derive the expected value of this process.
  - Derive the variance of this process.
  - Derive the covariance of this process for  $j = 1, 2, 3$ .
  - Derive the autocorrelation function of this process for  $j = 1, 2, 3$ .
4. Consider two separate stochastic processes:  $Y_t$  and  $X_t$ . Suppose that  $Y_t$  represents yearly excess returns and  $X_t$  represents yearly inflation. Suppose we are interested in eventually forecasting  $Y$  with the help of past values of  $Y$  as well as a past value of  $X$ . With this information, answer the following:
- Write down an ARMA(2,2) which also includes a (first) lagged value for  $X$ .
  - Write down the  $h$  - step ahead value for  $Y$  ( $Y_{t+h}$ ) that you put down in part (a).
  - Construct the forecast value of  $Y$  ( $\hat{Y}_{t+h|t}$ ) for  $h = 1$ .
  - Why is it difficult to forecast for  $h = 2$  and beyond?
5. In Los Angeles, researchers are examining the effect of air pollution on school attendance rates across various districts from 1970 to 1980, using a panel data model with time and school district effects.

- (a) Define fixed effects in a panel data model with time and school district effects. Write down the model that incorporates fixed effects for both time and school districts to investigate the relationship between air pollution levels and school attendance rates in Los Angeles during the 1970-1980 period.
  - (b) Discuss the advantages and limitations of employing school district fixed effects in analyzing the influence of air pollution on school attendance. How do these fixed effects help improve the model?
  - (c) What do the time fixed effects capture in this study?
  - (d) When estimating the impact of air pollution on school attendance with school district fixed effects, should researchers use cluster-adjusted standard errors? If so, briefly discuss the rationale and identify the appropriate level of clustering for this study.
6. You are given the regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

You want to test the null hypothesis

$$\begin{aligned} H_0 &: \beta_2 \beta_3 = \beta_1^3 \\ H_A &: \beta_2 \beta_3 \neq \beta_1^3 \end{aligned}$$

Show how you would construct the test statistic. Explicitly state the distribution and the degrees of freedom of your test statistic.

7. A local government in New York State implemented a new policy in 2010 that provides additional funding to students who score above a certain threshold on a standardized test. Given that the policy is intended to provide funding to students with test scores above the threshold, but some students below the threshold still receive funding due to local administrative decisions, and some students above the threshold do not receive funding, researchers are interested in studying the impact of this additional funding on future student performance using a regression discontinuity (RD) design.
- (a) Explain the difference between a sharp and a fuzzy regression discontinuity design. Clearly write the discontinuity condition for both types of RD designs.
  - (b) Determine which type of RD design (sharp or fuzzy) is more appropriate for this study. Justify your choice based on the variability in funding distribution described in the context.
  - (c) Does the RD model in this context lead to a simple two-stage least squares (2SLS) estimation strategy? If yes, write down the relevant equations and explain the first stage equation, the reduced form equation, the instrument, and the instrumented variables in your model. If no, explain why not.
8. A researcher is examining the impact of trade openness ( $S_i$ ) on economic growth ( $Y_i$ ) across different countries. To address potential endogeneity issues, the researcher uses the presence of a free trade agreement (FTA) with major trading partners as an instrumental variable ( $Z_i$ ). The researcher also includes control variables ( $X_i$ ), such as investment rates and political stability, to account for other factors that might influence economic growth.
- (a) Write the first stage and reduced form equations for the instrumental variables (IV) model, where the aim is to estimate the causal impact of trade openness ( $S_i$ ) on economic growth ( $Y_i$ ) using the presence of a free trade agreement ( $Z_i$ ) as the instrument, along with covariates ( $X_i$ ).
  - (b) Demonstrate that the IV estimator in your model can be expressed as a function of two coefficients obtained from the first stage and reduced form equations.
  - (c) Derive the reduced form equation for a Two-Stage Least Squares model by utilizing the first stage equation and the causal relation of interest mentioned above.
  - (d) What are the two key requirements for a good instrumental variable? Do you think the presence of a free trade agreement (FTA) with major trading partners satisfies these requirements? Explain your reasoning.